Pratibimb

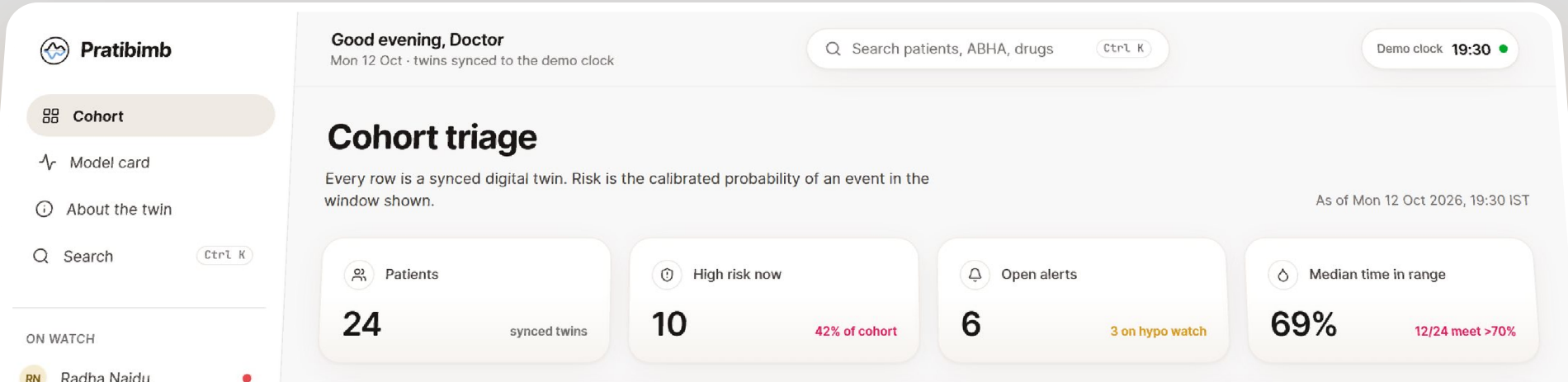
Happiest Health · Digital Twin Challenge 2026

प्रतिबिम्ब means “reflection” · a digital twin for Type 2 Diabetes

A living reflection of a person with diabetes.

A hybrid physiology + machine-learning twin, personalised from the health record, kept in sync with every CGM reading, and able to answer “what if?” before the patient tries it.

Team Stack Underflow · Yashvardhan Thanvi (lead, solo entry) · BIT Mesra, Off Campus Jaipur



Stack Underflow

A solo entry: one final-year BCA student, one digital twin.

YT

Yashvardhan Thanvi

Team leader

Final-year BCA
Birla Institute of Technology (BIT Mesra), Off Campus Jaipur

COLLEGE

Birla Institute of Technology (BIT Mesra)

Off Campus Jaipur, Rajasthan, India

Programme: Bachelor of Computer Applications (BCA), final year

CHALLENGE	CONDITION	WHAT WE BUILT	LICENCE
Happiest Health Digital Twin Challenge 2026 (Unstop)	Type 2 Diabetes, designed for India	Twin engine + risk models, FastAPI, clinician console, evaluation	Apache-2.0 · trained on synthetic data; validated on open real data

101 million Indians live with diabetes.

Between visits, no one sees the spike coming.

11.4%

~101 M adults in India live with diabetes

15.3%

~136 M more have prediabetes

35.5%

have hypertension

39.5%

have abdominal obesity

ICMR-INDIAB-17 national survey: Anjana RM et al., *Lancet Diabetes & Endocrinology* 2023.

SPARSE DATA

Glucose moves every few minutes; clinic labs arrive every few months. Wearables and CGM fill the gap, but nobody can watch the stream.

DIFFERENT PHENOTYPES

Asian Indian T2D splits into SIDD, SIRD, MARD and IROD subgroups, plus the "thin-fat" phenotype[2]. One model does not fit all.

DIFFERENT PLATES

Carbohydrate-heavy meals, evening chai and snacks, late dinners, festival sweets: the spikes follow the Indian day.

DIFFERENT RISKS

Sulfonylureas are widely prescribed. On a fasting day (Navratri, Ekadashi, Ramadan) a missed meal plus an SU dose is a classic hypoglycaemia trap.

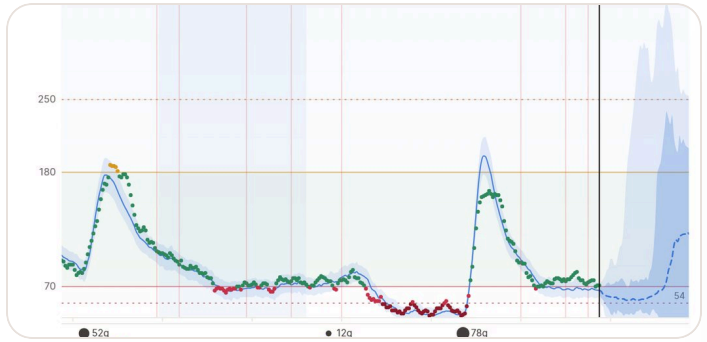
[2] INSPIRED cohort: Anjana RM et al., *BMJ Open Diabetes Research & Care* 2020.

A clinician following patients on CGM
needs to know who needs attention now, and why.



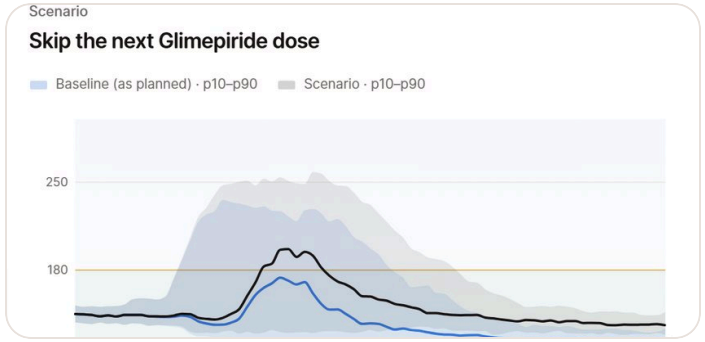
1 Triage

The cohort view ranks every synced twin by calibrated risk: level-2 hyperglycaemia within 2 h, hypoglycaemia within 1 h, nocturnal hypo tonight.



2 Understand

Open the virtual patient: CGM against the twin, a 4-hour forecast cone, and the reasons behind each risk (“time since sulfonylurea dose”).



3 Counsel

Test advice on this patient’s twin first: millet roti instead of rice, a walk after dinner, a skipped or shifted glimepiride dose, a short night.

THE BRIEF’S EXAMPLE

“Predict a sudden glucose spike 2 hours in advance” is our **hyper_2h** task. We add hypoglycaemia within 1 h and nocturnal hypoglycaemia tonight, because lows are what make sulfonylurea and insulin therapy dangerous.

Record, sense, mirror, foresee.

One twin per patient, updated every five minutes.

1 Record

The EHR (FHIR R4 with an ABHA-style ID) sets the twin's priors: HbA1c and fasting glucose for baseline levels, the TyG index for insulin resistance, diabetes duration for β-cell reserve.

2 Sense

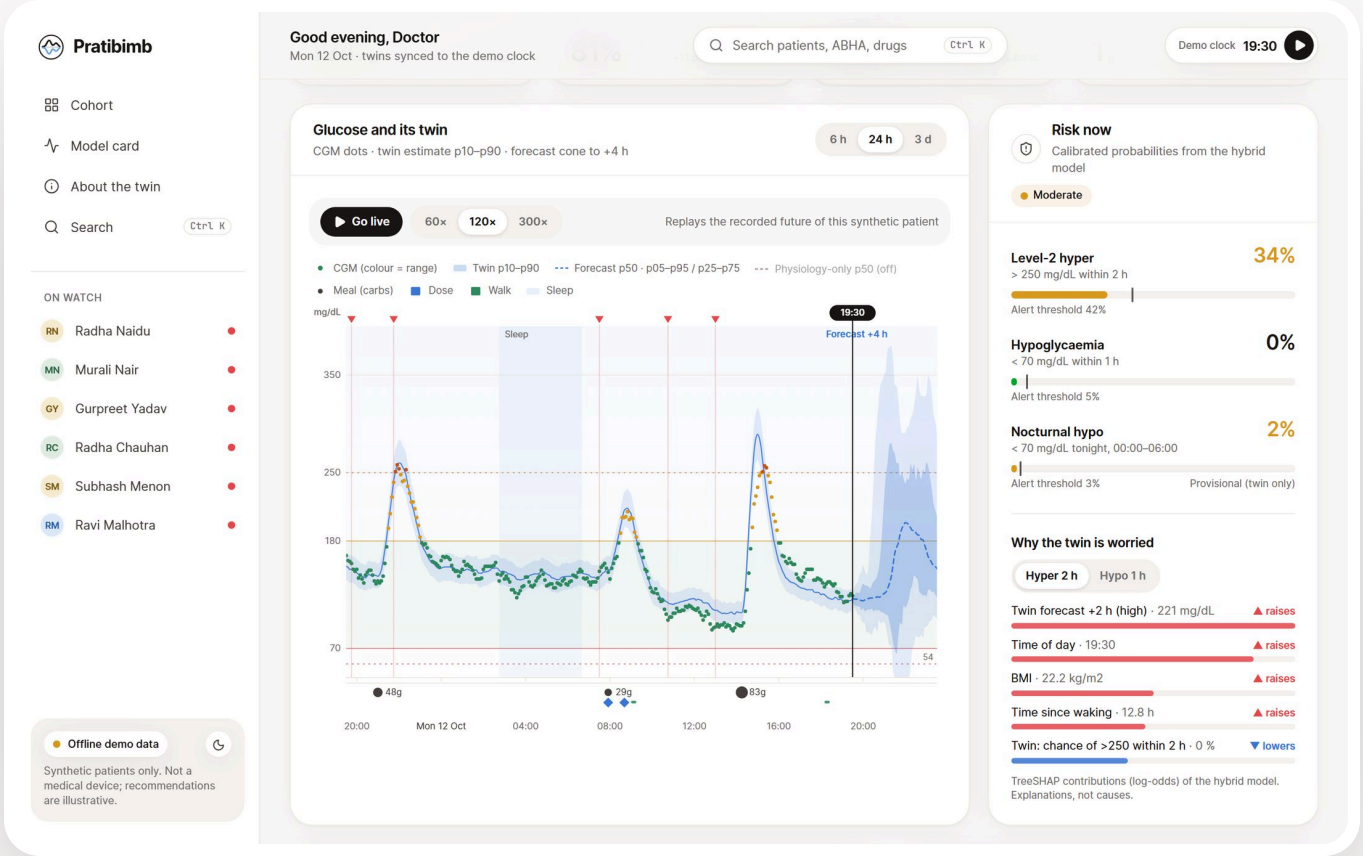
CGM every 5 minutes, heart rate, HRV, steps and sleep stages stream in with logged Indian meals and doses.

3 Mirror

A 64-member Ensemble Kalman Filter keeps a glucose–insulin model in sync, re-estimating insulin sensitivity, β-cell response and liver glucose output.

4 Foresee

The synced twin forecasts 4 hours ahead; a LightGBM layer turns it into calibrated, explained alerts and the What-If Lab tests advice.



The virtual patient in the clinician console: CGM dots, twin band (p10–p90), forecast cone to +4 h, calibrated risk with TreeSHAP reasons.

Not a classifier with a dashboard.
A model of this patient that stays in sync.

MECHANISTIC CORE

Physiology, not just patterns

An extended Bergman minimal model with gut absorption, exercise, sleep, stress, dawn effect and five Indian-common drug classes, personalised from the EHR.

CONTINUOUS SYNC

Re-estimated every 5 minutes

An Ensemble Kalman Filter updates hidden physiology from each CGM reading: insulin sensitivity, β-cell response, liver glucose output, meal absorption speed, dawn effect.

COUNTERFACTUALS

Answers “what if?”

The same synced ensemble simulates alternatives (meals, walks, doses, sleep) and returns baseline-vs-scenario bands, peaks, time above 180 / below 70 and hypo probability.

	RISK CLASSIFIER	PRATIBIMB
Personalised physiology (S_I, β, EGP)	no	**yes, with uncertainty**
Forecast with a calibrated cone	no	**p05–p95 to +4 h**
What-if on this patient	no	**meals · walk · dose · sleep**
Reasons a clinician can read	sometimes	**TreeSHAP + twin state**

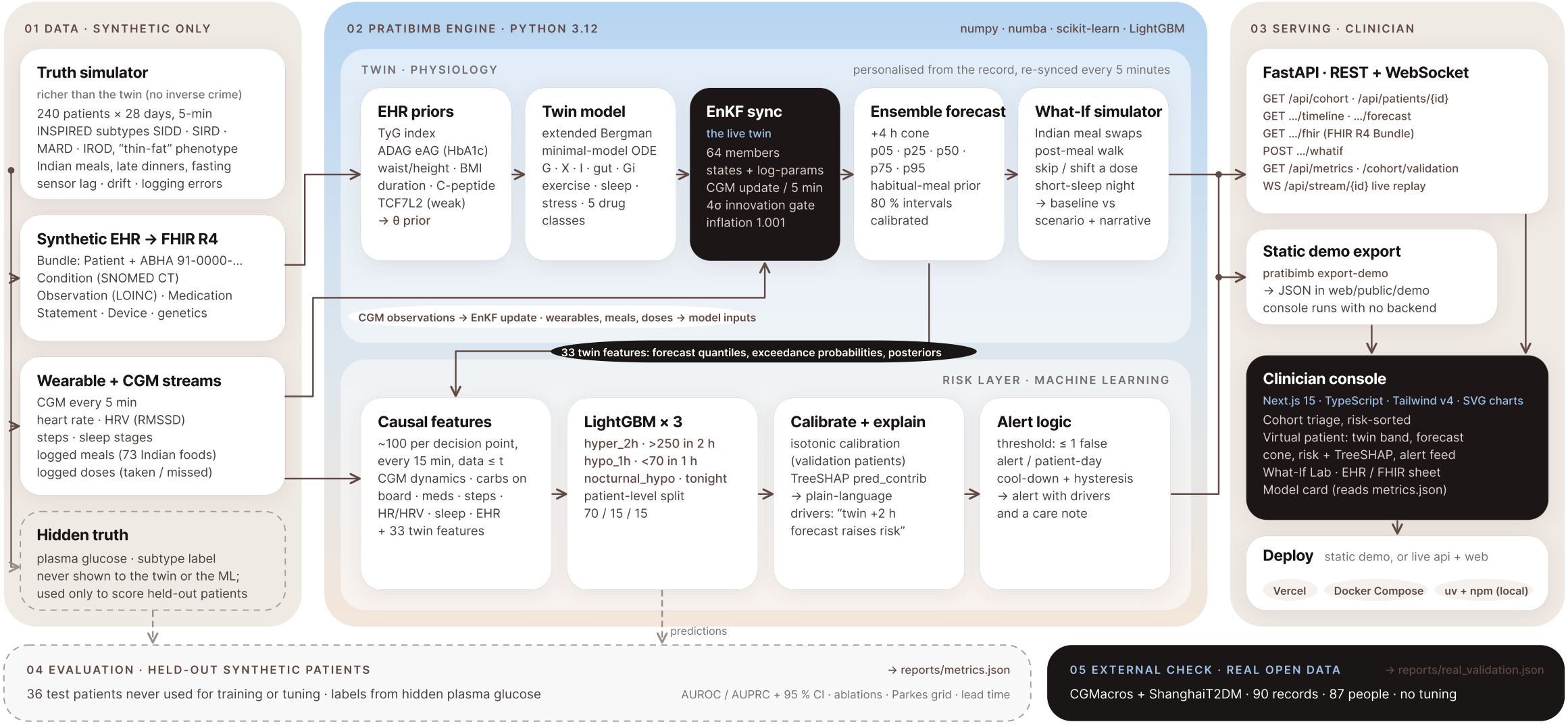
NO “INVERSE CRIME”

Test patients come from a **richer truth simulator** than the twin: drifting insulin sensitivity, illness, stress, unlogged snacks, CGM lag, drift and dropouts. The twin never sees true glucose or the subtype label.

SYSTEM ARCHITECTURE

Pratibimb · hybrid physiology + ML digital twin for Type 2 Diabetes

data / control flow · evaluation only (hidden truth) · core twin / UI · component



Training data are synthetic (DPDP Act / HIPAA sandbox); open de-identified real CGM data only for external validation. Not a medical device.

Synthetic by design, shaped like Indian EHR and wearable data.

STATIC · EHR

Demographics, 2 years of labs (HbA1c, FPG, lipids, creatinine/eGFR, ALT, UACR, C-peptide for ~35 %), comorbidities, medications, synthetic genetic markers (TCF7L2 rs7903146, PRS percentile).

FHIR R4 Bundle per patient: Patient with ABHA-format ID (`91-0000-…`, obviously synthetic), Condition (SNOMED CT), Observation (LOINC), MedicationStatement, Device.

DYNAMIC · WEARABLES

CGM every 5 min with lag, calibration drift, AR(2) noise, compression lows and dropouts; heart rate, HRV (RMSSD), steps, sleep stages; logged meals and doses with realistic logging errors.

240 × 28
patients × days, 5-min resolution

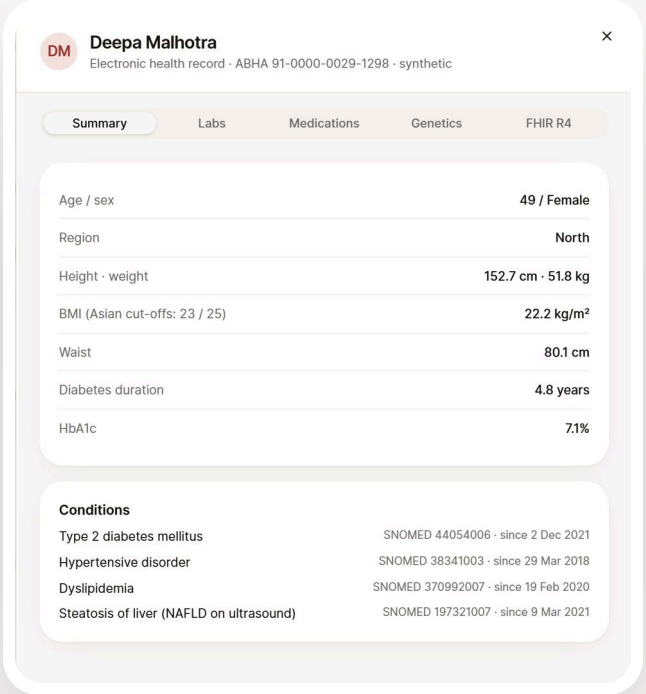
1,913,090
CGM points

168 / 36 / 36
train / val / test patients

65 · 33 · 46 · 96
SIDD · SIRD · MARD · IROD

REALISM CHECKS (COHORT_VALIDATION.JSON)

- Cohort-median TIR **68.7 %** (target ~55–75 %)
- Median CV **29.0 %**; 78.3 % of patients within 20–36 %
- **97 %** of hypo episodes in sulfonylurea / insulin users
- Mean CGM vs ADAG eAG from HbA1c: r = **0.914**



EHR · FHIR sheet in the console: summary, labs, medications, genetics and the raw FHIR R4 Bundle.

No real patient data in training | **DPDP Act 2023 / HIPAA-safe sandbox** | **Synthetic Indian names and ABHA-format IDs** | **Deterministic seed · rebuild with pratibimb build**

A compact physiological twin, personalised from the health record.

TWIN STATE-SPACE MODEL · 1-MIN STEP, 5-MIN OBSERVATIONS

$$dG/dt = -(S_G + X)\cdot G + EGP\cdot f_{EGP}(t) + Ra(t)/V_G - k_{ex}\cdot e(t)\cdot G - R(G)$$

$$dX/dt = -p_2\cdot X + p_2\cdot S_I\cdot m_{sleep}\cdot m_{stress}\cdot m_{ex}(t)\cdot I$$

$$dI/dt = -n\cdot I + \beta\cdot \max(G - h,\ 0)\cdot drug(t) + I_{exo}(t)/V_I$$

gut: $D_1 \to D_2 \to Ra$, τ from GI, fat, protein, fibre

$$dG_i/dt = (G - G_i)/\tau_i \quad (\text{CGM sees } G_i,\ \tau_i \approx 9 \text{ min})$$

- Drugs: metformin lowers EGP · sulfonylurea adds insulin secretion, part of it glucose-independent · DPP-4 boosts post-meal secretion · SGLT2i lowers the renal threshold 180 → 105 mg/dL · glargine as a slow depot
- Wearables: a short night lowers S_I by up to 20 % next day · stress (HR up, HRV down vs 7-day baseline) raises EGP · steps drive exercise uptake

EHR → PRIORS FOR Θ = {S_I, S_G, B, EGP, MEAL RATE, DAWN}

S_I insulin sensitivity	TyG index, waist/height, BMI
β β-cell response	duration, HbA1c, C-peptide, TCF7L2
EGP liver output	fasting glucose + ADAG eAG
S_G, meal rate, dawn	population priors

FORMULAS USED

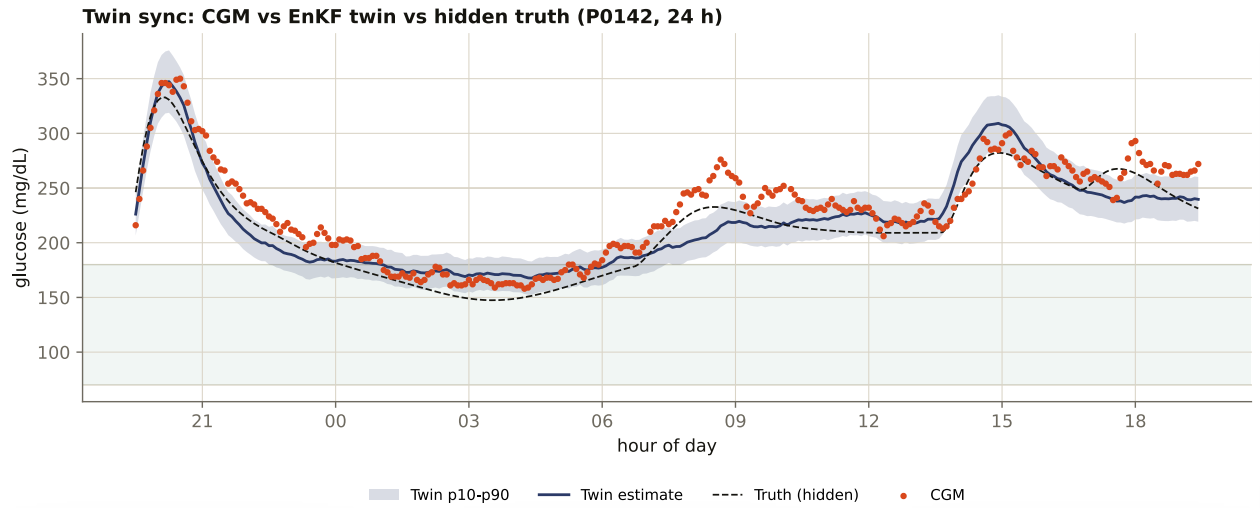
$$TyG = \ln(TG\ [mg/dL] \times FPG\ [mg/dL] / 2)$$

$$eAG\ (mg/dL) = 28.7 \times HbA1c - 46.7$$

The twin never reads the subtype label; the console shows an *inferred* subtype from twin posteriors + EHR. Full parameter table: docs/MODEL_CARD.md.

Kept in sync every five minutes
by an Ensemble Kalman Filter.

Twin sync on a held-out patient (P0142, 24 h): CGM, EnKF twin estimate with p10–p90 band, hidden truth



9.38 %
twin vs truth MARD

8.51 %
raw CGM vs truth MARD

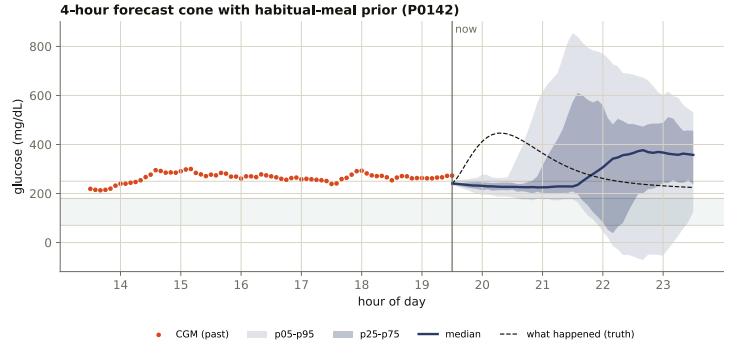
0.81–0.82
80 % forecast-interval coverage

Honest note: the filtered twin is slightly *less* accurate than raw CGM against truth (9.38 % vs 8.51 % MARD) because the EnKF cannot remove correlated sensor drift. Its value is in forecasting and in the features it feeds the risk models.

HOW THE SYNC WORKS

- 64 members; augmented state = 7 dynamic states + 6 parameters (log / logit space, slow random walk)
- Update on each CGM reading (obs. sd 5 + 6 %); missing CGM → no update
- 4σ innovation gate rejects compression artefacts; inflation 1.001
- Forecast 240 min with a habitual-meal prior; intervals widened by validation-fitted factors

4-hour forecast cone with habitual-meal prior (P0142)



Three early warnings, each with its reasons.

hyper_2h

Level-2 hyperglycaemia (>250 mg/dL for ≥15 min) begins within the next 120 min

hypo_1h

Hypoglycaemia (<70 mg/dL for ≥15 min) begins within the next 60 min

nocturnal_hypo

Any hypo between 00:00 and 06:00 tonight, predicted at bedtime

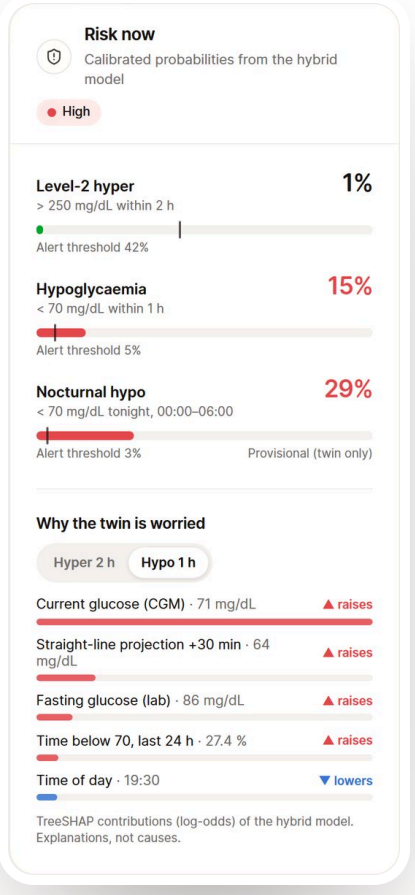
Labels come from hidden truth plasma glucose, never from CGM artefacts; a CGM-labelled variant is reported too.

PIPELINE (EVERY 15 MIN)

- ~100 strictly causal features (data ≤ t): CGM dynamics, carbs on board, meds, steps, HR/HRV, sleep, EHR, plus **33 twin features**
- LightGBM per task · isotonic calibration on validation patients
- TreeSHAP (`pred_contrib`) → plain-language drivers
- Alerts: threshold for ≤ 1 false alert per patient-day (nocturnal ≤ 1 per patient-week), cool-down + hysteresis

TOP TREESHAP DRIVERS (MEAN |CONTRIBUTION|)

hyper_2h	Twin: chance of >250 within 2 h · Time of day · Twin forecast +2 h (high)
hypo_1h	Current glucose (CGM) · Twin: forecast low in next 2 h · Time since sulfonylurea dose
nocturnal	Twin: expected overnight mean · Basal insulin action · Time since sulfonylurea dose

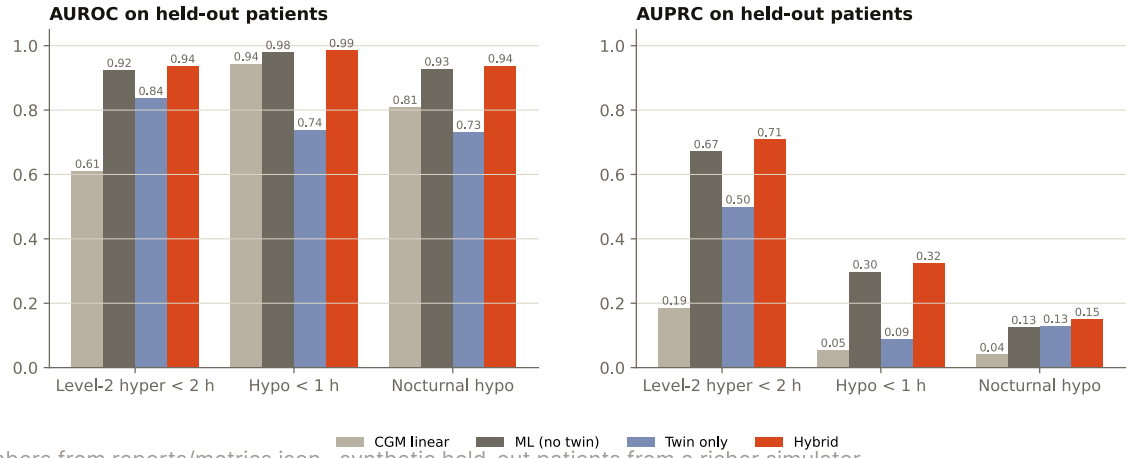


Radha Naidu (synthetic, P0098)

Scored on 36 patients
the models never saw.

TASK (HELD-OUT TEST)	PREVALENCE	AUROC (95 % CI)	AUPRC (95 % CI)	SENSITIVITY	PRECISION	ALERTS / PT-DAY	MEDIAN LEAD
hyper_2h	0.1337	**0.9368** 0.925–0.948	**0.7087** 0.641–0.765	0.6334	0.4686	2.0526	**80 min**
hypo_1h	0.0052	**0.9855** 0.977–0.993	**0.3244** 0.176–0.399	0.425	0.0794	0.6625	**35 min**
nocturnal_hypo	0.0085	**0.9379** 0.903–0.983	**0.1504** 0.054–0.232	0.875	0.0593	0.1261	**185 min**

Ablation on the same test patients: CGM linear · ML without twin · twin only · hybrid



THE TWIN EARNS ITS PLACE

Hybrid beats ML-without-twin on every task: hyper_2h AUROC 0.924 → **0.9368**, AUPRC 0.6717 → **0.7087**; hypo_1h AUPRC 0.298 → **0.3244**.

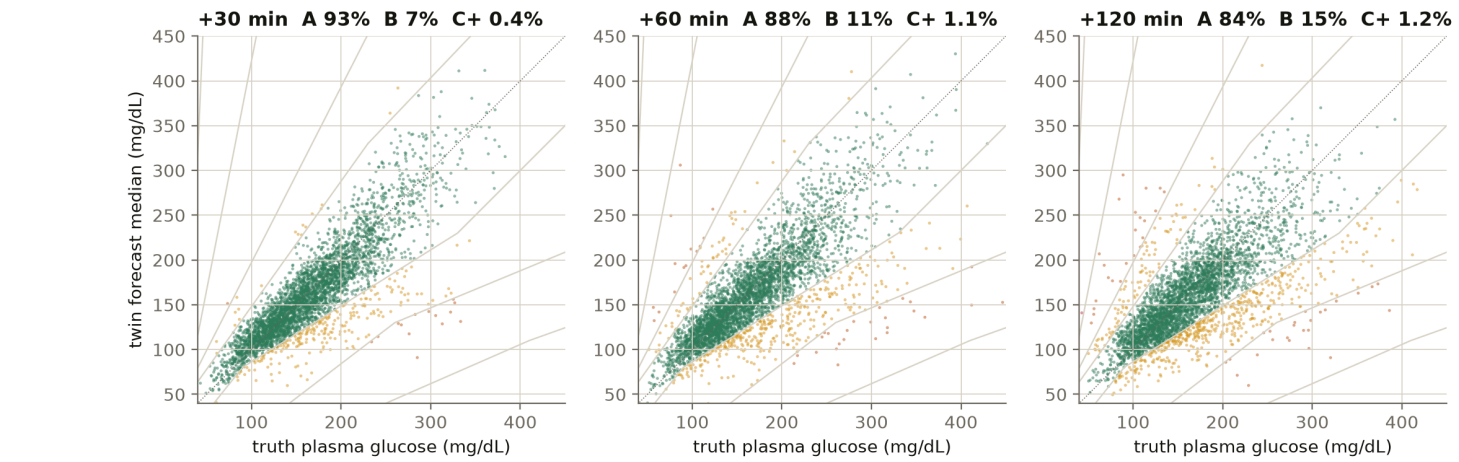
Operating points chosen on validation patients (≤ 1 false alert per patient-day). Sensitivity is per event; nocturnal rests on 8 positive test nights, so its CI is wide.

All numbers from reports/metrics.json · synthetic held-out patients from a richer simulator

Four-hour forecasts
that beat simple baselines.

HORIZON	TWIN RMSE (MG/DL)	MARD (%)	80 % INTERVAL COVERAGE	PARKES ZONE A (%)	LINEAR-EXTRAPOLATION RMSE	PERSISTENCE RMSE
30 min	**29.76**	11.89	0.8111	92.95	41.89	36.25
60 min	**37.81**	14.58	0.8151	88.14	76.51	49.52
120 min	**40.47**	16.7	0.8178	84.31	140.56	60.27

Parkes (consensus) error grid for type 2: twin forecast vs truth plasma glucose, held-out patients



0.00–0.01 %
of forecasts in Parkes zones D + E at any horizon; zone E 0.0 %

93,024
forecast points scored per horizon

Raw ensemble intervals were too narrow (coverage 0.36–0.43); widening by validation-fitted factors 3.103 / 2.991 / 2.783 gives the coverage shown.

All numbers from reports/metrics.json · synthetic held-out patients from a richer simulator

Validated on real patients

90 open CGM records, unchanged twin, no tuning.

REAL · CGMacros + ShanghaiT2DM (non-commercial licence)

CGMACROS · PHYSIONET

45

people · 45 records · 490 days · CC BY-NC-SA 4.0

Healthy, prediabetes and T2D. 5-min Dexcom G6, Fitbit heart rate and METs, photo-logged meal macros. No medication records.

SHANGHAIT2DM · FIGSHARE

42

people · 45 records · 561 days · CC BY 4.0

T2D, non-insulin records only. 15-min Libre, text meal records, oral agents mapped to Pratibimb drugs (glinide → short-acting SU; acarbose → slower absorption).

TWIN FORECAST ON REAL CGM

27.3 / 30.84 / 34.51

RMSE (mg/dL) at +30 / 60 / 120 min, 89,977 timestamps

- Beats linear extrapolation at every horizon (23.23 / 45.96 / 90.61).
- Beats persistence at 2 h (34.51 vs 40.6); loses at 30–60 min.
- Synthetic-fitted 80 % intervals still cover 0.7818 / 0.7636 / 0.7586.

Real data enter only as an external check. GlycoTwin's harmonised store of two open, de-identified datasets is read once; derived records stay in a git-ignored folder (CGMacros is CC BY-NC-SA 4.0) and only aggregate metrics are published. Twin Parkes zone A: 88.78 / 84.6 / 80.25 %.

Same timestamps, same folds, against a data-driven model.

FORECAST RMSE (MG/DL) · 89,977 IDENTICAL REAL TIMESTAMPS

HORIZON	TWIN	PERSISTENCE	LINEAR	GLYCOTWIN LIGHTGBM	TWIN MARD %	TWIN 80 % COV.	TWIN PARKES A %
30 min	**27.3**	18.52	23.23	14.06	13.6	0.7818	88.78
60 min	**30.84**	28.83	45.96	21.7	15.39	0.7636	84.6
120 min	**34.51**	40.6	90.61	27.87	17.22	0.7586	80.25

RISK AUROC / AUPRC · GLYCOTWIN FEATURES, PATIENT-LEVEL 5-FOLD CV

EVENT	BASE RATE	WINDOWS	BASELINE	+ TWIN FEATURES	SYNTHETIC MODEL, NO RETRAINING
≥180 mg/dL within 2 h	0.1304	78,970	0.8681 / 0.5341	**0.8703 / 0.5366**	0.6628 / 0.2363
>250 mg/dL within 2 h	0.0246	88,585	0.9232 / 0.3552	**0.9235 / 0.3554**	0.7902 / 0.1543

GlycoTwin LightGBM = its own out-of-fold predictions at the same timestamps (trained on all 154 records with patient-level CV). Risk models use fixed LightGBM settings; nothing was tuned on test folds.

What the real data told us, and what we change next.

FINDINGS

- A data-driven forecaster beats the physiological twin at every horizon (14.06 / 21.7 / 27.87 vs 27.3 / 30.84 / 34.51 mg/dL).
- Twin features add no measurable discrimination on real data (≥180: 0.8681 → 0.8703; >250: 0.9232 → 0.9235 AUROC).
- The synthetic-trained model transfers poorly (AUROC 0.7902 for >250, 0.6628 for ≥180).
- Our synthetic cohort is too sick: 2.479× the real ≥180 event rate, 5.439× the >250 rate and 1.532× the 30-min volatility.

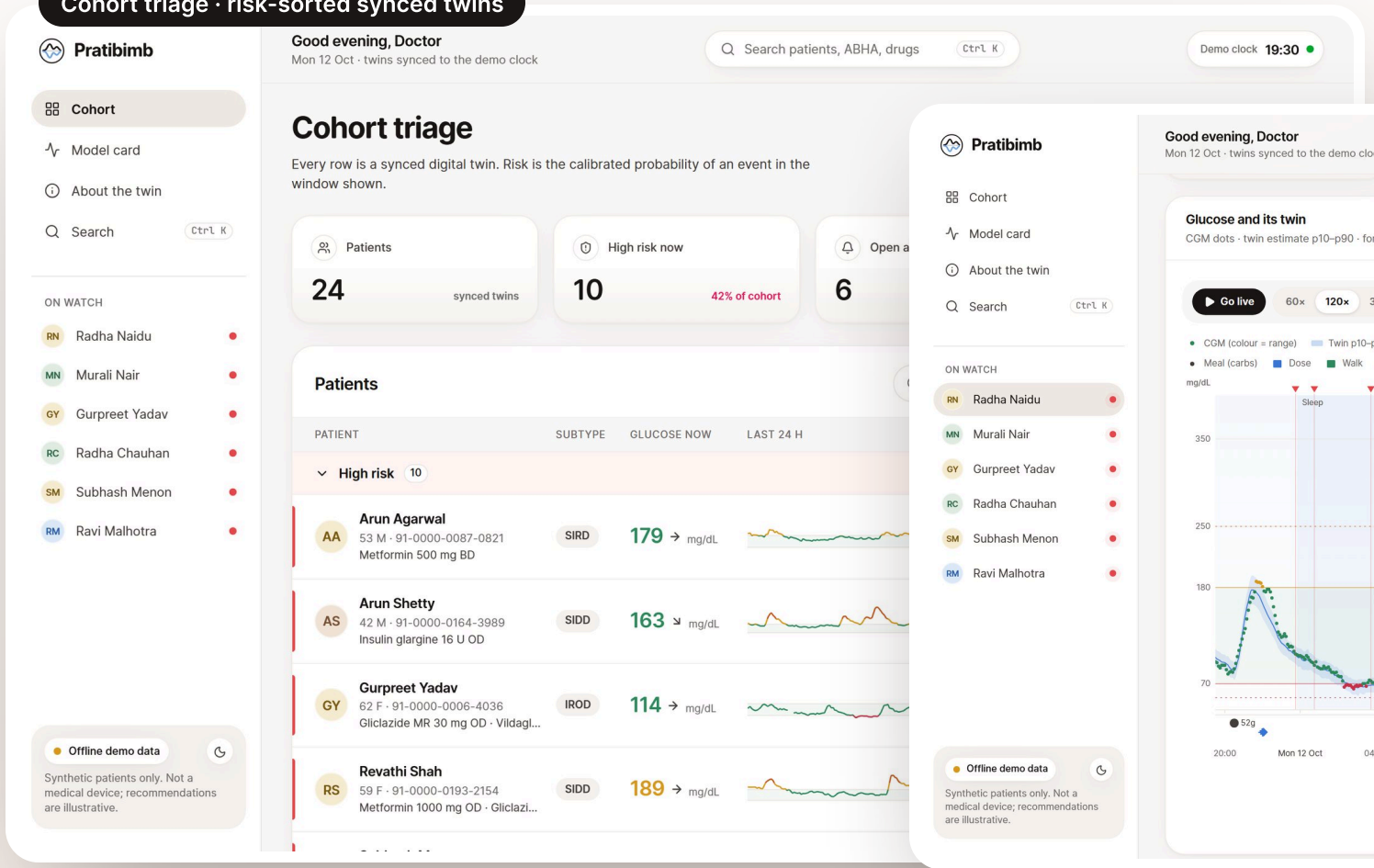
CAVEATS & NEXT STEPS

- 87 people from China and the USA, not India; insulin users excluded; no waist, sleep or HRV; meals are photo or text estimates.
- Recalibrate the simulator to real event rates and volatility, then retrain.
- Anchor short horizons to the CGM and blend the twin with a data-driven forecaster.
- Prospective validation on Indian patients before any clinical claim.

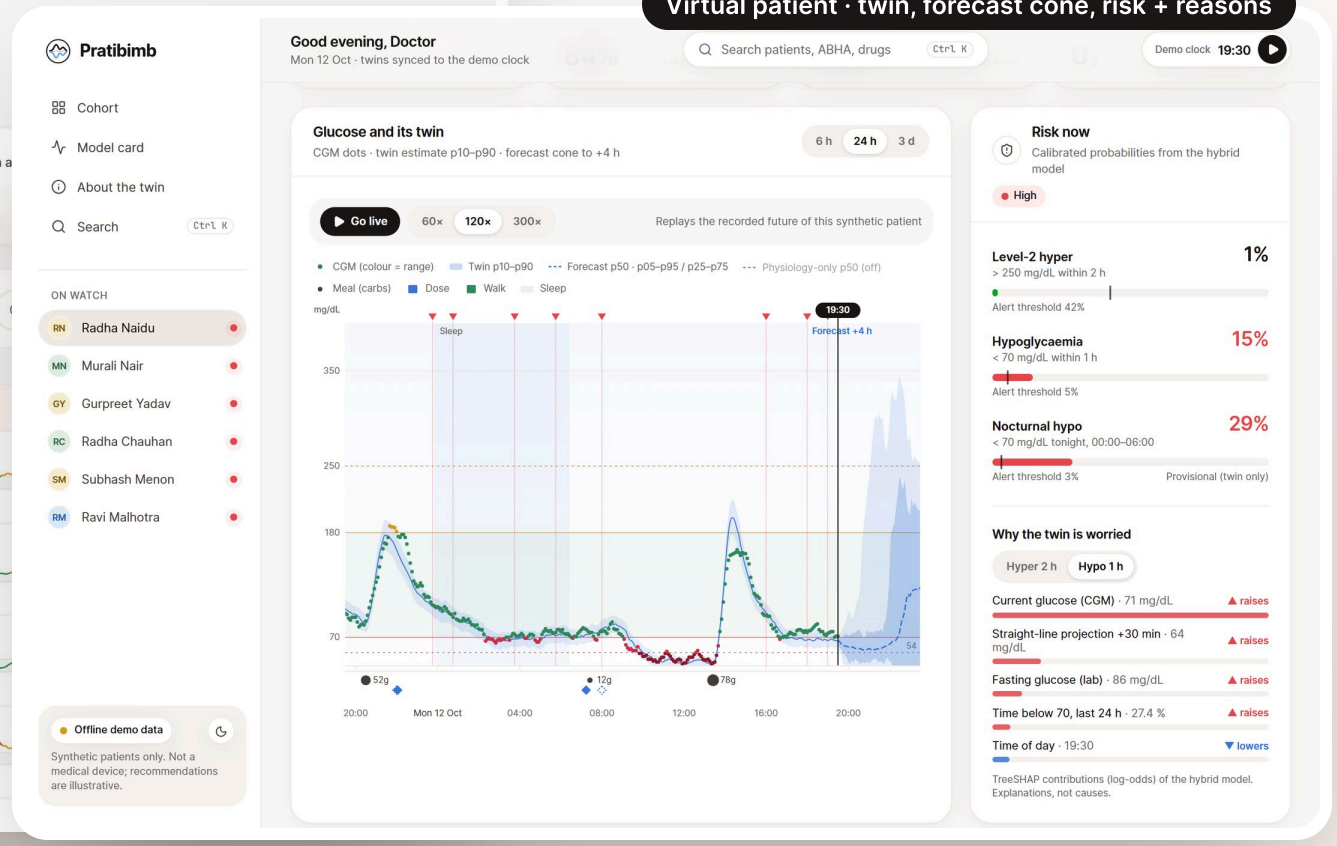
A console that feels like a product, built for a 30-second triage.

Live replay over WebSocket · EHR · FHIR sheet · AGP + TIR report · Inside the twin · Model card · Offline demo mode

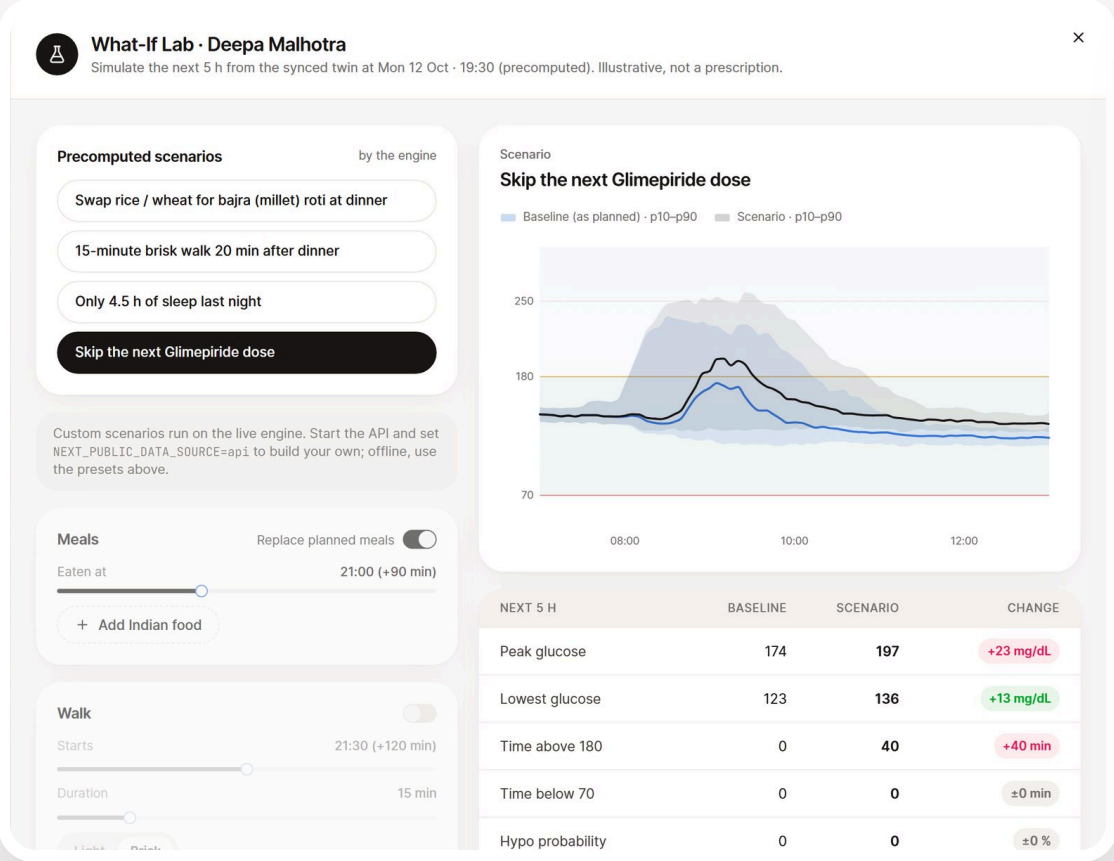
Cohort triage · risk-sorted synced twins



Virtual patient · twin, forecast cone, risk + reasons



Try the advice on the twin
before the patient tries it.



Deepa Malhotra (synthetic, P0029): skip the next glimepiride dose, baseline vs scenario bands.

ENGINE PRESETS (DEMO EXPORT, NEXT 5 H)

Skip next glimepiride · P0029

Peak 174 → 197 mg/dL **+23** · time >180: 0 → 40 min · TIR 100 % → 89 %

15-min brisk walk after dinner · P0029

Peak 204 → 199 mg/dL **−5** · time >180: 65 → 55 min · TIR 78 % → 82 %

Bajra (millet) roti at dinner · P0098

Peak 166 → 150 mg/dL **−16**

Skip next glimepiride · P0098 (hypo-prone)

Chance of a low 89 % → 83 % · time <70: 90 → 80 min · peak 131 → 158 mg/dL

Food picker over 73 Indian foods with Hindi names · walk · skip/shift a dose · last night's sleep. Narratives are computed from the deltas (no LLM). Illustrative, not prescriptions.

Built for India, not translated to it.

ABDM-READY RECORDS

Every synthetic patient is an HL7 **FHIR R4** Bundle, the format of the Ayushman Bharat Digital Mission, with an ABHA-format identifier, LOINC labs and SNOMED CT conditions.

```
"code": {"coding": [{"system":
"http://loinc.org",
 "code": "4548-4"}]}  // HbA1c
```

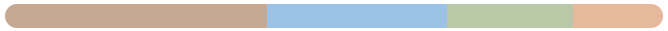
73 INDIAN FOODS

Katori / roti / piece portions with Hindi names, carbs, fibre, GI and GL from IFCT 2017 and the International GI Tables, with an honest source column.

सफ़ेद चावल · GI 73 · बाजरे की रोटी · GI 55 · पोहा · GI 70 · इडली · GI 69 · राजमा · GI 24 · चाय (चीनी वाली)

INSPIRED SUBTYPES

Cohort archetypes from Asian Indian T2D subgroups: IROD ~39 %, SIDD ~26 %, MARD ~22 %, SIRD ~13 %, including the “thin-fat” phenotype and Asian BMI cut-offs (23 / 25 kg/m²).



Cohort: IROD 96 · SIDD 65 · MARD 46 · SIRD 33

SULFONYLUREA SAFETY

Fasting days with an SU dose are simulated. 97 % of hypo episodes occur in SU / insulin users, and “time since sulfonylurea dose” is a top driver of hypo alerts. The What-If Lab shows what skipping or shifting a dose does.

LATE DINNERS & CHAI

Late dinners (≈ 20:00–22:45), evening chai + snack, festival sweets, region-flavoured meal templates. Median post-meal rise: lunch 122.1, dinner 111.8, breakfast 95.5 mg/dL.

ROUTINE LABS ONLY

Fasting insulin is rarely measured in India, so insulin resistance comes from the **TyG index** (triglycerides × fasting glucose), and baseline glucose from HbA1c via ADAG.

What we do not claim, and how we keep it safe.

LIMITATIONS

- Trained on synthetic data only. The real-data check shows synthetic accuracy does not carry over; prospective validation is required before any claim.
- The truth simulator is richer than the twin but from the same minimal-model family; real physiology will be further away.
- Posterior S_I and β are “effective” values: use them as trends, not lab measurements.
- Hypo alerts have low precision (0.0794 for hypo_1h) because lows are rare; nocturnal metrics rest on 8 positive nights.
- Food GI values are approximate; meal logging is assumed timely.

SAFETY & ETHICS

- Not a medical device. Alerts and recommendations are illustrative; the clinician decides.
- Training data are synthetic (names, ABHA-format IDs); real data only as open, de-identified datasets for an external check (DPDP Act 2023 / HIPAA sandbox).
- TreeSHAP drivers explain the model, not the patient’s biology: “explanations, not causes”.
- The inferred subtype agrees with the hidden archetype 50.8 % of the time: a rough phenotype, never a diagnosis.
- No LLM in the loop: forecasts, risks and what-if narratives are deterministic and reproducible.

INTENDED USE

A conceptual clinician console for triage and counselling, demonstrated on synthetic patients for this challenge.

OUT OF SCOPE

Any clinical decision or dosing, identifiable patient data, paediatric use, type 1 diabetes, pregnancy.

From proof of concept to a clinic pilot.

Phase 1

Validate

Re-run evaluation on open T2D CGM datasets and de-identified clinic data under ethics approval; recalibrate thresholds per site.

Phase 2

Integrate

ABDM sandbox with consent-based FHIR R4 exchange; CGM and wearable vendor feeds; clinician sign-in.

Phase 3

Pilot

Clinician-in-the-loop study at a diabetes clinic: alert burden, time to action, usefulness of what-if counselling.

Phase 4

Extend

Patient app in Hindi and regional languages; hypertension and CKD co-twins; lighter on-device sync.

POTENTIAL IMPACT, IF VALIDATED

Between-visit care that sees what is coming

- Earlier action on spikes and lows between clinic visits
- Safer sulfonylurea use around fasting days
- Counselling grounded in the patient's own physiology and plate
- ABDM-ready records and open-source code from day one

AVAILABLE TODAY

Engine + tests (pytest) · FastAPI · REST + WebSocket · Clinician console · Static demo export · Docker Compose · Model & data cards

Pratibimb

Thank you.

धन्यवाद

Video walkthrough (≥ 20 min)	**https://youtu.be/3gu_ek68Mm0**
Live demo	**https://pratibimb-omega.vercel.app**
Source code	**https://github.com/Thanatos9404/pratibimb**
Docs in the repo	README · docs/MODEL_CARD.md · docs/DATA_CARD.md · docs/API.md

Team Stack Underflow · Yashvardhan Thanvi (lead, solo entry) · BIT Mesra, Off Campus Jaipur

Apache-2.0 · trained on synthetic data, validated on open real data · not a medical device · sky photos: Unsplash (credited in web/public/photos/CREDITS.md)